Cainiao is Fast — But Not Everywhere

When does Alibaba's logistics network stop delivering its speed advantage?

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Abstract

Using 1.29 million Tmall orders (Jan–Jul 2017), we test whether Alibaba's Cainiao logistics platform makes order fulfillment faster, and whether that advantage is uniform. Cainiao is about nine hours faster on average, and the gap survives a fair-comparison control. But a hierarchical breakdown reveals a sharp exception: for big-city destinations handled by small carriers, Cainiao is actually *slower* than non-Cainiao fulfillment, customer review scores are lowest, and the bottleneck sits in between-facility line-haul rather than the last mile. We translate this into three concrete recommendations for Cainiao. **Audience: Cainiao (the logistics platform).**

1 Introduction and Motivation

In Chinese e-commerce, delivery speed is one of the most visible dimensions of competition and a direct driver of sales and customer loyalty. Fisher et al. (2019) show, in a quasi-experiment at a U.S. apparel retailer, that online sales rise by about 1.45% for every business-day reduction in delivery time; Rao et al. (2011) show the converse—failing to deliver on the promised date erodes the future purchasing of previously loyal customers. Speed is therefore not a vanity metric: it maps to revenue and retention.

Cainiao's core value proposition is that its smart-routing platform makes fulfillment faster than carriers acting alone. In our data Cainiao handles **30.2%** of all orders, so any *systematic* pocket where that promise fails is material to the platform and to the merchants who rely on it. This study asks a single, focused question and pursues it in depth rather than reporting many shallow comparisons.

Research question. *Does Cainiao make order fulfillment faster, and is that advantage uniform across carriers and destinations?*

2 Data and Method

Data. We use three linked files covering Jan 1–Jul 31, 2017: an *order* table (1,291,197 orders: payment time, amount, Cainiao flag, review, promised-speed window), a *logistics* event log (969 MB: per-order milestone scans `CONSIGN` → `GOT` → `ARRIVAL/ DEPARTURE` → `SENT_SCAN` → `SIGNED`, with facility, city, and carrier IDs), and a *merchant* table (daily visitors and running review scores). This is a real-world dataset with missing milestone entries for some orders.

Outcome. Total fulfillment time follows the course definition:

$$\text{totaltime} \;=\; \text{SIGNED timestamp} \;-\; \text{pay\_timestamp} \quad \text{(hours)}.$$

We keep the 1,288,438 orders (99.8%) with a valid, positive total time below a 30-day cap.

Hierarchical design. Rather than running unrelated parallel comparisons, we dig one level at a time, adding a variable only after the previous result motivates it (Table 1): (i) Cainiao vs. non-Cainiao; (ii) + carrier size; (iii) + destination-city size; (iv) a segment decomposition that localizes the bottleneck. Group cut-offs follow the starter analysis: a carrier is **Large** if it handles >200,000 orders (the two largest carriers dominate volume), and a destination is a **Big city** if it is one of the two largest signing cities.

Table 1: Key constructed variables.

Variable	Definition
`totaltime`	SIGNED − pay (hours)
`pre_delivery`	GOT − pay (order → picked up)
`line_haul`	SENT_SCAN − GOT (between-facility transit)
`last_mile`	SIGNED − SENT_SCAN (final station → buyer)
`LCsize`	Large if carrier handles >200k orders, else Small
`Citysize`	BigCity if destination is a top-2 signing city, else SmallCity
`n_city`	distinct cities a package passes through

3 Descriptive Overview

Before the hierarchy, two facts frame the analysis. First, fulfillment is slow on average—the median order takes 64 h (2.7 days) from payment to signature—so hours saved are economically meaningful given the speed–sales link above. Second, the Cainiao speed gap is not a one-month artifact: plotting daily average fulfillment time over the whole Jan–Jul 2017 window (Figure 1) shows Cainiao orders running consistently below non-Cainiao orders across the entire period, including the volume-heavy spring months. This temporal stability is what licenses pooling the data for the cell-level comparisons that follow. The Tableau view plots the daily group mean: blue represents Cainiao and orange represents non-Cainiao.

4 Findings

4.1 Overall, Cainiao is faster

Cainiao orders are fulfilled in **61.9 h** on average versus **71.0 h** for non-Cainiao—about **9.2 hours faster** (medians 53.7 vs. 66.3 h; Figure 2). A common confound is that Cainiao might simply handle orders with no tight promise. Restricting to orders with *no* promised-speed window (`promise` = 0) leaves the gap essentially unchanged (62.5 vs. 71.0 h), so the advantage is not an artifact of promise mix (Table 2). In the Tableau comparison, bar height is mean fulfillment time; blue is Cainiao and orange is non-Cainiao.

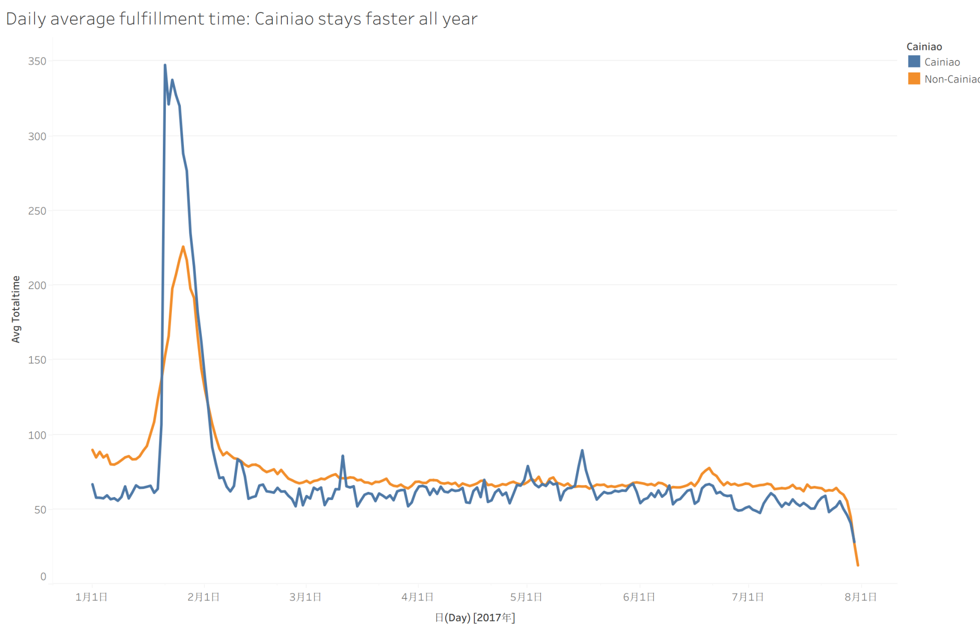

Figure 1: Daily average fulfillment time by group (Tableau); the Cainiao advantage is stable over time.

Table 2: L1 main effect and the no-promise control.

Group	n orders	Avg time (h)	Median (h)
Cainiao	388,144	61.9	53.7
Non-Cainiao	900,294	71.0	66.3
Control: promise $= 0$ *only*			
Cainiao	374,836	62.5	—
Non-Cainiao	900,257	71.0	—

4.2 The advantage is uneven across carriers

Splitting by carrier size, Cainiao's edge is larger on large carriers (60.3 vs. 70.3 h, a 10.0 h saving) than on small ones (66.3 vs. 73.3 h, 7.0 h; Table 3). The advantage shrinks but does not yet disappear— which motivates adding the destination dimension.

Table 3: L2: average fulfillment time (hours) by Cainiao $\times$ carrier size.

Carrier	Group	n orders	Avg time (h)
Large	Cainiao	287,405	60.3
Large	Non-Cainiao	666,479	70.3
Small	Cainiao	100,739	66.3
Small	Non-Cainiao	233,815	73.3

4.3 Punchline: big-city destinations on small carriers

Adding destination-city size exposes the real story (Table 4, Figure 3). Cainiao saves 8–10.5 hours in three of the four cells— but in **big-city destinations handled by small carriers, Cainiao**

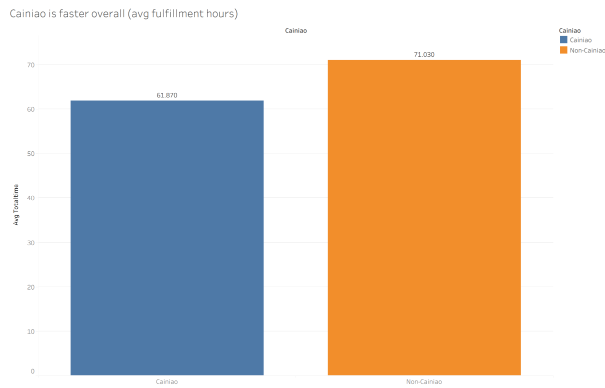

Figure 2: Cainiao is faster overall (Tableau).

is 3.3 hours *slower* than non-Cainiao. The cell is not thin: it contains 22,756 orders, so the reversal is not a small-sample fluke. The Tableau bars show mean fulfillment hours within each destination-size/carrier-size cell, with blue for Cainiao and orange for non-Cainiao.

Table 4: L3 (punchline): average time, sample size, and Cainiao's advantage by cell.

Destination	Carrier	Cainiao (h)	Non-Cainiao (h)	**Cainiao saved (h)**	n (both)
Big city	Large	44.7	54.2	**+9.5**	96,301
Big city	Small	52.4	49.1	**−3.3**	22,756
Small city	Large	61.7	72.2	**+10.5**	857,583
Small city	Small	67.1	75.2	**+8.0**	311,798

4.4 Customer reviews echo the punchline

The same cell is Cainiao's weakest on logistics review score: 4.764 for Cainiao— its lowest of any cell—versus 4.860 for non-Cainiao, the highest of any cell (Figure 5). Customers feel the slowdown, so this is an experience problem, not only an operational curiosity. The Tableau view reports the mean review score in each cell; blue denotes Cainiao and orange denotes non-Cainiao.

Data quality and missingness. As a real-world feed, the logistics log is incomplete for some orders: a small share lack one or more milestone scans (`GOT`, `SENT_SCAN`, or `SIGNED`), which is why our valid sample is 1.288 M of 1.291 M orders and why segment means are computed over the subset of orders that carry the relevant pair of scans. We made two deliberate choices to keep conclusions honest: (i) the total-time analysis uses only orders with a genuine `SIGNED`–payment span (positive, under a 30-day cap), excluding implausible records; and (ii) segments are reported as directional diagnostics rather than an exact additive decomposition, since their subsets differ slightly from the totals. Because missingness is spread across groups rather than concentrated in the reversal cell, it cannot manufacture the punchline.

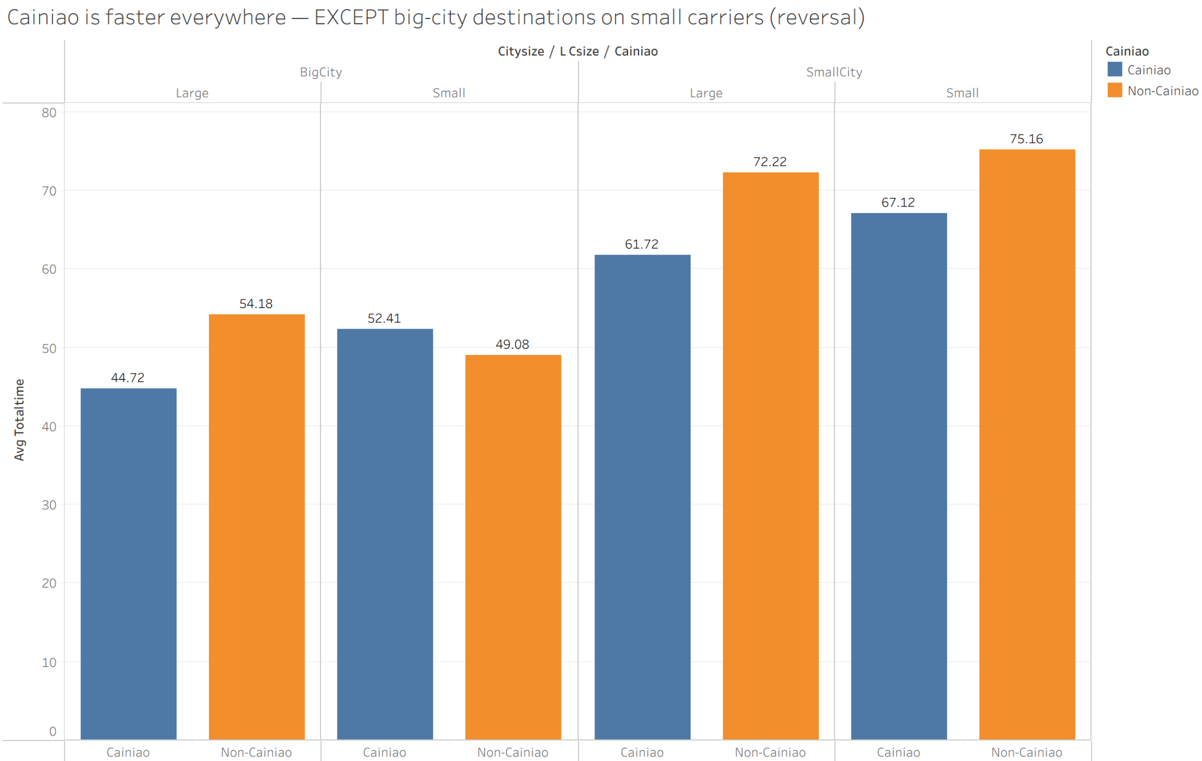

Figure 3: Tableau: average fulfillment time by destination-city size × carrier size, split Cainiao vs. non-Cainiao. In three cells the Cainiao (blue) bar is shorter; in big-city/small-carrier the Cainiao bar is *taller*—the reversal.

5 Mechanism: where does the time go?

Decomposing fulfillment time into pre-delivery, line-haul, and last-mile localizes the problem (Table 5, Figure 6). Last-mile is small and similar across all groups (roughly 4–6 h), so it is *not* where the gap lives. Cainiao's usual edge comes from much faster between-facility line-haul (e.g. 38.4 vs. 47.8 h on small-city/large-carrier orders). In the punchline cell that edge disappears: Cainiao line-haul is 27.0 h versus non-Cainiao 26.9 h, essentially tied—so the lost overall advantage is a lost *line-haul* advantage. In the Tableau stacked bars, red is pre-delivery, orange is line-haul, and blue is last-mile; each segment height is the mean hours for that cell.

Table 5: Average segment time (hours) by cell. Segment means use orders with complete milestones and are not strictly additive to Table 4.

Destination	Carrier × Group	Pre-delivery	Line-haul	Last-mile
Big city	Small, Cainiao	0.36	27.0	3.67
Big city	Small, Non-Cainiao	0.29	26.9	4.66
Small city	Large, Cainiao	0.30	38.4	5.31
Small city	Large, Non-Cainiao	0.32	47.8	5.26
Small city	Small, Non-Cainiao	0.30	51.0	6.11

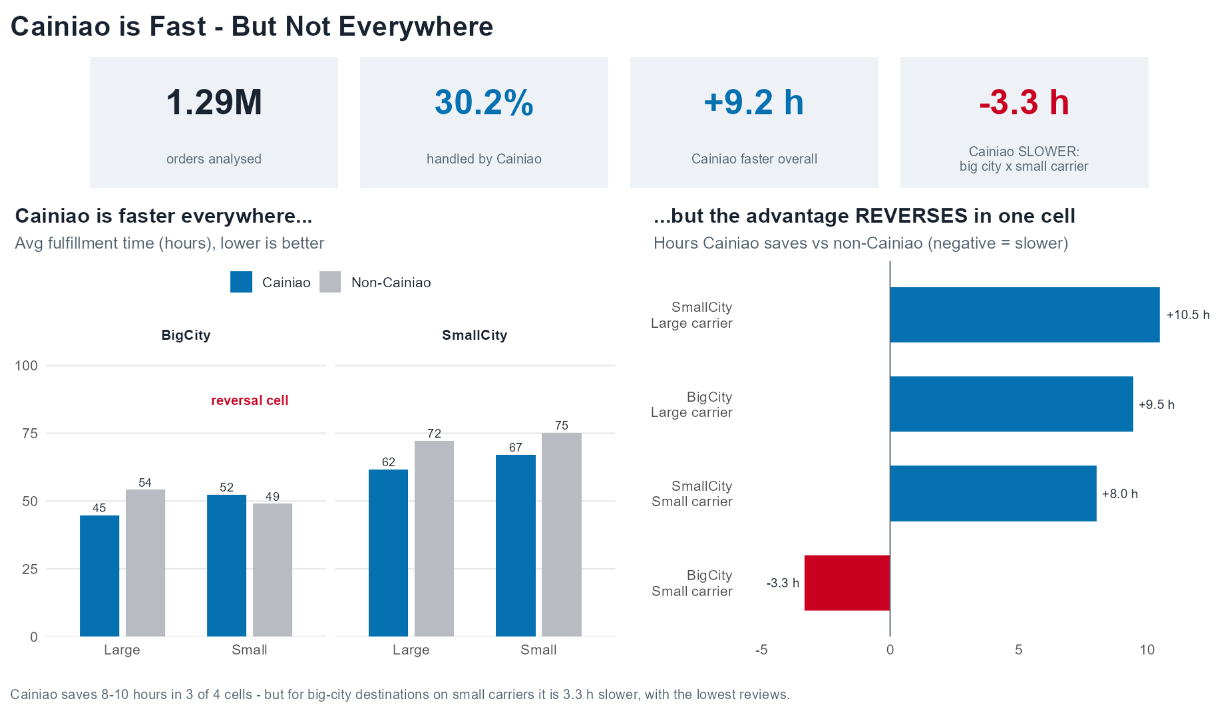

Figure 4: Composite decision dashboard (R): the KPI strip, the 2×2 breakdown, and Cainiao's hours-saved per cell in one view; the big-city/small-carrier cell is the only one where Cainiao saves negative hours (−3.3 h, red).

An honest non-result. A natural hypothesis—that small carriers make more consolidation stops—is *not* supported. Small carriers pass through slightly *fewer* cities on average (2.9–3.1) than large carriers (3.4–3.7). The penalty is therefore per-leg transit and dwell time, not a larger number of hops. We report this explicitly because it disciplines the explanation we offer.

5.1 Robustness of the cut-offs

The punchline does not hinge on arbitrary thresholds. The carrier-size split is economically grounded: order volume is highly concentrated, with the two largest carriers covering roughly three-quarters of all orders, so the "Large vs. Small" boundary separates network-scale incumbents from the long tail rather than slicing a continuum at a fragile point. The destination split likewise isolates the two dominant signing cities, which carry distinct trunk-line economics. Crucially, every cell in Table 4 is populated with tens of thousands of orders—the smallest, the reversal cell, still holds 22,756—so none of the comparisons rests on thin data. The reversal is also corroborated by an independent signal (customer reviews) and a mechanism (line-haul), which a spurious threshold effect would not jointly produce.

6 Potential Reasons

Two domain explanations are consistent with the evidence. First, big cities concentrate parcel volume and inter-city trunk lines; large carriers run dedicated, high-frequency line-haul into them, while small carriers must wait to consolidate loads, adding transit and dwell precisely on the high-

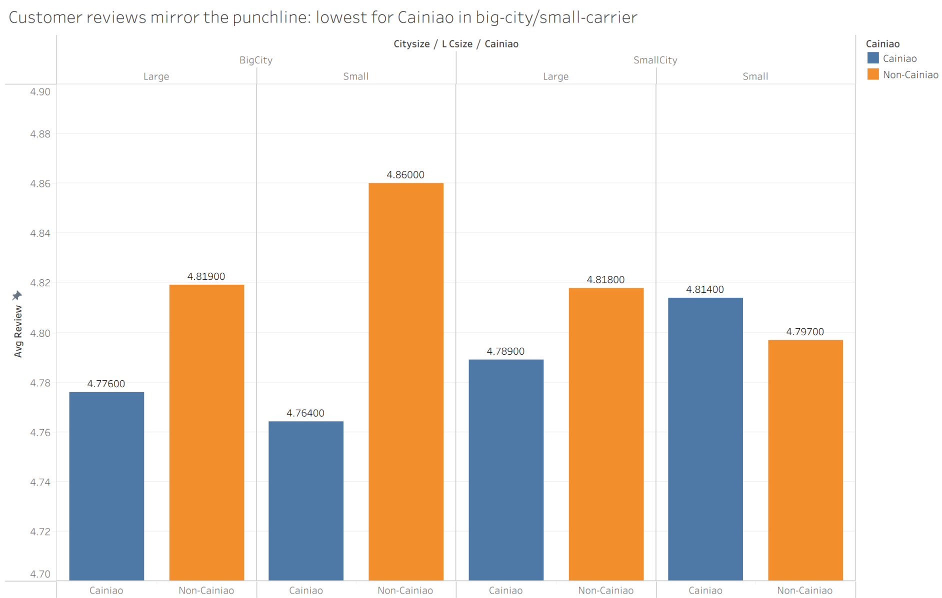

Figure 5: Customer reviews mirror the speed punchline (Tableau).

expectation big-city lanes. This matches the practice view of Chinese e-commerce logistics, in which network density and trunk-line scale are decisive for speed (Yu et al., 2016). Second, small carriers likely lack the capacity to execute Cainiao's smart-routing recommendations as designed; the heterogeneous-fleet routing literature shows that ignoring vehicle/carrier heterogeneity degrades realized routing performance (Koç et al., 2016), so a one-size-fits-all optimization need not translate into time saved for resource-constrained carriers.

7 Practical Implications (So What?)

For Cainiao, three actions follow directly from the hierarchy:

1. **Deepen relationships with large carriers on big-city lanes**, where they reliably convert into speed—while monitoring the concentration risk this dependence creates.
2. **Make smart-routing carrier-capability-aware** (size, coverage, experience), so small carriers are not assigned routes they cannot execute efficiently (Koç et al., 2016).
3. **Pilot big-city consolidation hubs and parcel lockers for small-carrier shipments**, attacking the line-haul and pickup penalty where it actually occurs; parcel-locker networks are an established lever for last-mile and collection efficiency (Deutsch and Golany, 2018).

Sizing the opportunity. The reversal cell is small in share but not negligible in absolute terms: the 5,720 Cainiao orders in big-city/small-carrier lanes each run about 3.3 h slower than the non-Cainiao alternative and 7.7 h slower than Cainiao's own big-city/large-carrier performance (44.7 h). Closing even half of that gap would remove on the order of tens of thousands of order-hours of delay per the seven-month window—and, given the speed–sales elasticity in Fisher et al. (2019), a measurable sales effect—while also lifting the lowest review scores in the network. Because the weakness is concentrated, the fix is cheap relative to a platform-wide intervention.

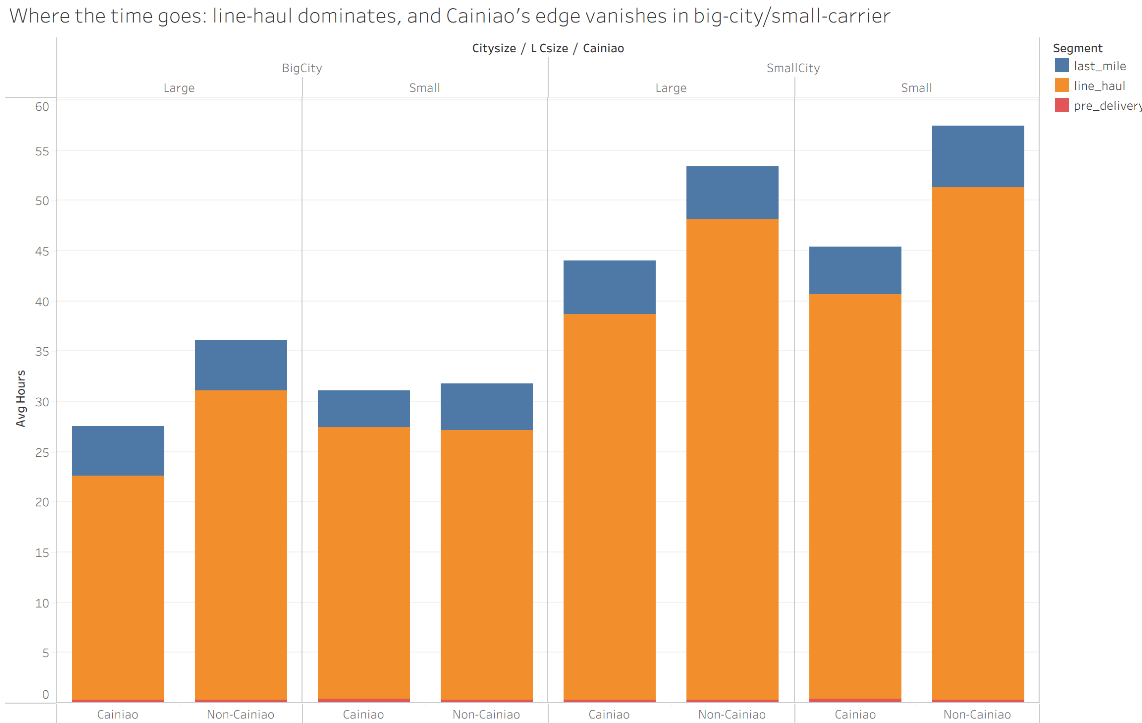

Figure 6: Segment decomposition (Tableau): stacked pre-delivery / line-haul / last-mile per cell and group. The line-haul leg dominates and is where Cainiao normally wins; that win evaporates for big-city/small-carrier orders.

8 Decision Support: an Interactive Story

The findings are packaged as a Tableau story built directly on the exported aggregates (see `tableau/TABLEAU_BUILD_GUIDE.md`). It walks a Cainiao stakeholder from the headline (Cainiao is faster) through the hierarchical breakdown (the big-city/small-carrier cell highlighted in a contrasting color), the segment decomposition (line-haul, not last mile), and the review echo, ending on the three recommendations. The design keeps a single accent color for Cainiao and greys out non-focus marks so the reversal cell carries the eye—turning a multi-table analysis into a one-screen narrative a manager can act on.

9 Limitations and Causality

This is observational data: assignment to Cainiao is not random, so differences are associations, not proven causal effects—merchants who choose Cainiao may differ systematically (e.g. product mix, geography). Milestone records are missing for some orders, so segment means use slightly different subsets than the totals; we therefore treat segments as directional, not additive. The big-city flag is a coarse two-city proxy, and review scores are dense near the ceiling. These caveats temper the magnitudes but not the qualitative punchline, which is large and consistent across time, customer reviews, and the segment decomposition.

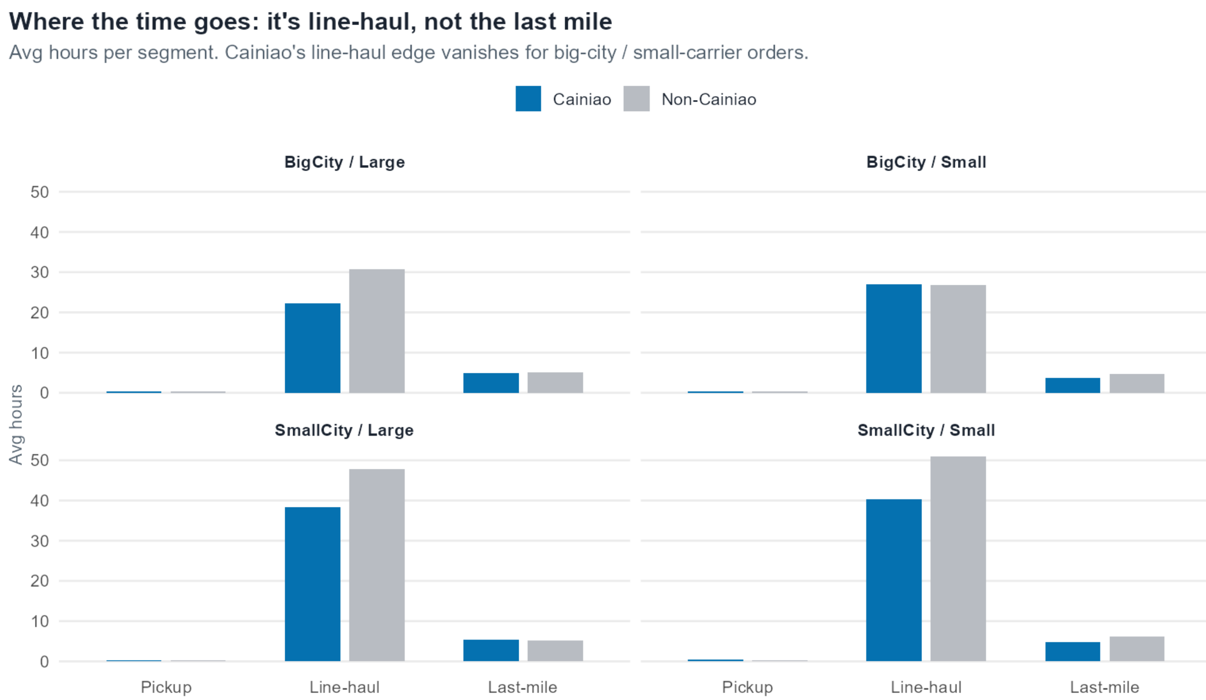

Figure 7: Composite mechanism dashboard (R): per-segment averages faceted across the four cells. Last-mile is small and flat; the gap lives in line-haul, which stays high for big-city/small-carrier Cainiao orders.

10 Conclusion

(1) Cainiao significantly shortens order fulfillment time across the marketplace—about nine hours on average, robust to a promise control. **(2)** But that advantage largely disappears—and even reverses—for big-city destinations served by small carriers, with the bottleneck in between-facility line-haul rather than the last mile. The implication for Cainiao is targeted, not wholesale: fix the specific carrier-by-destination pocket where the platform's promise currently breaks.

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